## Executive Summary

NYC 311 Operational Analytics | 2024

This project analyzes New York City's 2024 311 service requests to evaluate service demand, agency workload, resolution performance, and data quality. Using SQL for data preparation and analysis and Power BI for visualization, the project transforms a large operational dataset into insights that can support more informed resource planning and service-delivery decisions.

## Key Performance Indicators

| KPI | 2024 result |
|---|---|
| Total service requests | 3M |
| Closure rate | 98.88% |
| Average resolution time | 13.75 days |
| Median resolution time | 0.34 days |
| Non-closed status requests | 39K |
| Requests missing closure dates | 63K |

## Key Findings

1. Service demand patterns

Brooklyn recorded the highest service-request volume, while Staten Island recorded the lowest. Heat and Hot Water requests increased during winter, indicating the importance of seasonal planning for housing-related service demand.

2. Agency workload and operational efficiency

NYPD accounted for the largest share of agency workload. The Department of Parks and Recreation had the longest average resolution time among agencies in the comparison. Staten Island also had the longest average resolution time among boroughs. These differences warrant further investigation, taking request type and case complexity into account.

3. HPD complaint resolution

Among the HPD complaint types analyzed, Unsanitary Condition had the longest average resolution time. This can help inform more focused monitoring of complaint categories with longer resolution cycles.

4. Data quality

The analysis identified requests marked Closed without a closure date, as well as non-closed requests that nevertheless contained a closure date. Approximately 2,991 non-closed requests had a closure date. These inconsistencies can affect the reliability of resolution-time reporting and should be validated before operational decisions are made.

**Executive recommendations**

- Use borough-level demand patterns to support workload allocation and resource planning.
- Prepare for increased winter demand in Heat and Hot Water complaints.
- Review long-resolution cases by agency, borough, and complaint type to understand whether delays relate to case complexity, operational capacity, or other factors.
- Strengthen validation between request status and closure dates.
- Monitor both average and median resolution times. The substantial difference between 13.75 days and 0.34 days suggests that the average may be influenced by a smaller number of long-running cases.

**Conclusion**

The NYC 311 Operational Analytics project demonstrates how SQL and Power BI can be used together to examine public-service demand, operational performance, and data reliability. The findings highlight differences across boroughs, agencies, and complaint categories while identifying data-quality issues that may affect performance measurement. The resulting dashboard provides a foundation for more consistent monitoring, evidence-based resource planning, and improved operational reporting.